

HDOC – Experiment 15

Question

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- 1. Check whether all digits are even.**
- 2. Check whether all digits are prime.**

❖ Understanding problem

Input

- Positive Integer
- User's Menu Choice

Output

Option 1

- All digits are even
- All digits are not even

Option 2

- All digits are prime
- All digits are not prime

❖ Test case table

TEST CASE	INPUT (choice, number)	EXPECTED OUTPUT	ACTUAL OUTPUT	RESULT
1	1, 24680	All digits are even	Same	Pass
2	1, 12345	All digits are not even	Same	Pass
3	2, 2357	All digits are prime	Same	Pass
4	2, 2379	All digits are not prime	Same	Pass
5	1, 888888	All digits are even	Same	Pass

❖ Algorithm

1. Start.
2. Display the menu with two options.
3. Read the user's choice.
4. Read the positive integer.
5. Copy the number into a temporary variable.
6. According to the selected option:-
 - Case 1: Check each digit.
 - If any digit is odd, display All digits are not even.
 - Otherwise, display All digits are even.
 - Case 2: Check each digit.
 - Prime digits are 2, 3, 5, and 7.
 - If any digit is not one of these, display All digits are not prime.
 - Otherwise, display All digits are prime.
7. If the menu choice is invalid, display an error message.
8. Stop.

❖ Program code in C

```
#include <stdio.h>

int main()
{
    int choice;

    long long number, temp;

    int digit;

    int allEven = 1;

    int allPrime = 1;


    printf("==== MENU =====\n");

    printf("1. Check Whether All Digits are Even\n");
    printf("2. Check Whether All Digits are Prime\n");


    printf("Enter your choice: ");

    scanf("%d", &choice);


    printf("Enter a positive integer: ");

    scanf("%lld", &number);


    temp = number;


    switch(choice)
    {
        case 1:

            while(temp > 0)
```

```
{  
    digit = temp % 10;  
  
    if(digit % 2 != 0)  
    {  
        allEven = 0;  
        break;  
    }  
  
    temp = temp / 10;  
}  
  
if(allEven)  
    printf("\nAll digits are even.\n");  
else  
    printf("\nAll digits are not even.\n");  
  
break;
```

case 2:

```
while(temp > 0)  
{  
    digit = temp % 10;  
  
    if(!(digit == 2 || digit == 3 || digit == 5 || digit == 7))  
    {  
        allPrime = 0;
```

```
        break;
    }

    temp = temp / 10;
}

if(allPrime)
    printf("\nAll digits are prime.\n");
else
    printf("\nAll digits are not prime.\n");

break;

default:

    printf("\nInvalid Choice!\n");
}

return 0;
}
```

❖ Compile, Execute and Evaluate

Test case 1

```
(kali㉿kali)-[~/Desktop/exp15]
$ ./digit_check
===== MENU =====
1. Check Whether All Digits are Even
2. Check Whether All Digits are Prime
Enter your choice: 1
Enter a positive integer: 24680

All digits are even.
```

Test case 2

```
(kali㉿kali)-[~/Desktop/exp15]
$ ./digit_check
===== MENU =====
1. Check Whether All Digits are Even
2. Check Whether All Digits are Prime
Enter your choice: 1
Enter a positive integer: 12345

All digits are not even.
```

Test case 3

```
(kali㉿kali)-[~/Desktop/exp15]
$ ./digit_check
===== MENU =====
1. Check Whether All Digits are Even
2. Check Whether All Digits are Prime
Enter your choice: 2
Enter a positive integer: 2357

All digits are prime.
```

Test case 4

```
(kali㉿kali)-[~/Desktop/exp15]
$ ./digit_check
===== MENU =====
1. Check Whether All Digits are Even
2. Check Whether All Digits are Prime
Enter your choice: 2
Enter a positive integer: 2379

All digits are not prime.
```

Test case 5

```
(kali㉿kali)-[~/Desktop/exp15]
$ ./digit_check
===== MENU =====
1. Check Whether All Digits are Even
2. Check Whether All Digits are Prime
Enter your choice: 1
Enter a positive integer: 888888

All digits are even.
```

❖ Compilation and Execution screenshot

```
(kali㉿kali)-[~/Desktop/exp15]
$ vi digit_check.c

(kali㉿kali)-[~/Desktop/exp15]
$ gcc digit_check.c -o digit_check

(kali㉿kali)-[~/Desktop/exp15]
$ ./digit_check
===== MENU =====
1. Check Whether All Digits are Even
2. Check Whether All Digits are Prime
Enter your choice: 1
Enter a positive integer: 24680

All digits are even.

(kali㉿kali)-[~/Desktop/exp15]
$ ./digit_check
===== MENU =====
1. Check Whether All Digits are Even
2. Check Whether All Digits are Prime
Enter your choice: 1
Enter a positive integer: 12345

All digits are not even.

(kali㉿kali)-[~/Desktop/exp15]
$ ./digit_check
===== MENU =====
1. Check Whether All Digits are Even
2. Check Whether All Digits are Prime
Enter your choice: 2
Enter a positive integer: 2357

All digits are prime.
```

```
(kali㉿kali)-[~/Desktop/exp15]
$ ./digit_check
===== MENU =====
1. Check Whether All Digits are Even
2. Check Whether All Digits are Prime
Enter your choice: 2
Enter a positive integer: 2379

All digits are not prime.

(kali㉿kali)-[~/Desktop/exp15]
$ ./digit_check
===== MENU =====
1. Check Whether All Digits are Even
2. Check Whether All Digits are Prime
Enter your choice: 1
Enter a positive integer: 888888

All digits are even.
```

